Harsha S. Bhat

Directeur de recherche CNRS & Professeur attaché
École Normale Supérieure, Paris

Laboratoire de Géologie,
24 Rue Lhomond, 75005 Paris, France
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Scholar ZHskR34AAAAJ

EDUCATION

2021	**H.D.R.**, *Supershear Earthquakes* — École Normale Supérieure, France †
2007	**Ph.D.**, Mechanical Sciences — Harvard University, USA *
2002	**M.S.**, Engineering Sciences — Harvard University, USA
2001	**B.E.**, Civil Engineering — NITK, India

† Habilitation à Diriger des Recherches *∗ Supervised by Prof. James R. Rice & Dr. Renata Dmowska*

CURRENT POSITIONS

2024	**Directeur de recherche CNRS** — École Normale Supérieure, France
2026	**Professeur attaché** — École Normale Supérieure, France

PAST POSITIONS

2022 – 2026	Professeur chargé de cours — École Polytechnique, France
2016 – 2024	Chargé de recherche CNRS — École Normale Supérieure, France
2021 – 2023	Visiting Professor — NISER, India
2012 – 2016	Chargé de recherche CNRS — Institut de Physique du Globe de Paris, France
2010 – 2011	Assistant Professor (Research) — University of Southern California, USA
2007 – 2010	Post-Doctoral Fellow — University of Southern California, USA
2007 – 2010	Visitor in Aeronautics — California Institute of Technology, USA
2007 – 2007	Post-Doctoral Fellow — Harvard University, USA
2001 – 2007	Graduate Research Associate — Harvard University, USA

FUNDING & GRANTS

∗ → waiting for results *gray → unsuccessful*

2026∗	*Agence Nationale de la Recherche – PRCI , France-HongKong* *“REFRESH: Revisiting Seismic Hazard and Fault Mechanics in Intraplate Regions under Climatic Stresses”*	coPI
2026∗	*Partenariat Hubert Curien PROCOPE , France–Germany* *“CASPER: Climate-Aware Seismic Hazard from Physics-Based Earthquake-Cycle Models”*	coPI
2026	*Domaine d’Intérêt Majeur MaTerRE , Île-de-France* *“GEO-PROBE: GPU-Accelerated Multi-Physics Spectral Boundary-Element Modeling of Induced Seismicity during Underground Gas Storage”*	coPI
2024	*Laboratoire de Recherche Conventionné (CEA) , France* *“CLIMSEISM: Climate-Triggered Intraplate Seismicity — Detection and Mechanics”*	coPI
2023	*Agence Nationale de la Recherche (AAPG 2022) , France* *“SMEC: Smectite Dehydration — a Key to Fault Mechanics”*	coPI
2023	*Partenariat Hubert Curien PROCORE , France–HongKong* *“Earthquake Risks along the Anninghe Fault Zone”*	coPI
2022	*Partenariat Hubert Curien Aurora , France–Norway* *FARM: Fault and Rupture Mechanics*	coPI
2022	*CNRS – University of Tokyo , France–Japan* *“SESAME: Seasonal Seismicity — Statistical Analysis and Modeling”*	coPI

2021	*Centre National d'Études Spatiales (CNES), France* *On the Link between Rupture Velocity and Off-Fault Damage Produced in Earthquakes*	coPI
2020	*Endeavour Fund (MBIE), New Zealand* *Can Earthquake Simulators Forecast Seismic Hazard after the Next Great Alpine Fault Earthquake?*	coPI
2019	*European Research Council, Consolidator Grant, EU* *"PERSISMO: Predicting Energy Release in Fault Systems — Integrating Simulations, Machine Learning and Observations"*	PI
2019	*Institut National des Sciences de l'Univers (CNRS), TelluS-ALEAS, France* *IRA: Inverting Rupture Velocity Using Damage and Aftershocks*	PI
2019	*Marie Skłodowska-Curie ITN (H2020), EU* *FRICFAULT: Friction in Faults*	coPI
2019	*Émergence (Ville de Paris), France* *Inelastic Deformation in the Subduction Zone*	coPI
2019	*Laboratoire de Recherche Conventionné (CEA), France* *"Earthquake Dynamics, Damage & Tsunamigenesis"*	coPI
2019	*Aramont Fund (Harvard), USA* *"Resolving Earthquake Source Mechanisms on Complex Fault Systems (fellow: K. Okubo)"*	coPI
2019	*Scheme for Promotion of Academic and Research Collaboration (SPARC), India* *"Novel Explorations on Earthquake Nucleation within Complex Fault Networks"*	coPI
2018	*PRESTIGE Grant, France* *"Seismic Hazard on Thrust Faults Using Physics-Based Models and Surface-Data Analysis (fellow: L. Bruhat)"*	host
2018	*Thomas Jefferson Fund (FACE Foundation), France–USA* *Earthquakes in Complex Fault Networks*	coPI
2018	*École Normale Supérieure (Actions Incitatives), France* *"Earthquake Mechanics"*	PI
2017	*Fondation Simone et Cino Del Duca (Académie des sciences), France* *Dynamics of Fault Networks: Physics-Based Models*	PI
2017	*Institut National des Sciences de l'Univers (CNRS), mi-lourds, France* *Observational and Computational Earthquake Cycles*	PI
2017	*Thomas Jefferson Fund (FACE Foundation), France–USA* *Earthquakes in Complex Fault Networks*	coPI
2016	*Agence Nationale de la Recherche, France* *LABSEISM: Laboratory Investigation of Earthquakes Triggered by Seismic Waves*	PI
2016	*Institut National des Sciences de l'Univers (CNRS), TelluS-ALEAS, France* *"Spontaneous Off-Fault Damage Due to Earthquakes and the Associated Acceleration Spectra"*	PI
2016	*Institut de Physique du Globe de Paris (Mobilité), France* *"FDEM Rupture Modeling at Los Alamos (PhD mobility: K. Okubo)"*	coPI
2015	*Agence Nationale de la Recherche, France* *LABSEISM: Laboratory Investigation of Earthquakes Triggered by Seismic Waves*	coPI
2015	*Émergence (Ville de Paris), France* *DEED: Deep Earthquake Dynamics*	coPI
2015	*Natural Environment Research Council (Fellowship), UK* *Seismic and Aseismic Slip: Properties and Constitutive Behavior of Faults*	coPI
2015	*Université Sorbonne Paris Cité, France* *"Earthquake Ruptures on Complex Fault Systems and Complex Media (PhD fellowship K. Okubo)"*	PI
2014	*European Research Council, Starting Grant (H2020), EU* *UNITERRA: Unifying On- and Off-Fault Earthquake Models*	PI
2014	*Émergence (Ville de Paris), France* *Quantifying the Seismic Hazard Associated with Waste-Water Injection*	PI
2014	*Natural Environment Research Council, UK* *Earthquake Fracture Damage and Feedbacks in the Seismogenic Zone*	coPI
2014	*Agence Nationale de la Recherche, France* *LIVOGERM: Lithospheric Volcanism and Geothermal Energy at Major Strike-Slip Fault Intersections*	coPI

2014	*Université Sorbonne Paris Cité , France* *LABSEISM: Laboratory Investigation of Earthquakes Triggered by Seismic Waves*	coPI
2013	*European Research Council, Starting Grant , EU* *SEISMECH: Towards the Next Generation of Mechanical Models of Earthquake Source Physics*	PI
2013	*Émergence (Ville de Paris) , France* *GEOSEISMECH: Integrating Mechanical Models with Geological, Experimental and Seismological Observations*	PI
2013	*AXA Research Fund , France* *"Seismic Hazard of the Shallow Portion of Megathrusts (postdoc: N. Cubas)"*	host
2013	*Southern California Earthquake Center (SCEC) , USA* *"Dynamic Formation of Fault-Zone Rocks and Their Influence on Rupture Propagation"*	PI
2012	*Agence Nationale de la Recherche (Chaires d'Excellence) , France* *GEOSEISMECH: Geological and Seismological Mechanics*	PI
2012	*Agence Nationale de la Recherche (Blanc, SIMI 6) , France* *GeoSMEC: Geometry of Strike-Slip Faults through Multiple Earthquake Cycles*	PI
2011	*Émergence (Université Pierre et Marie Curie) , France* *From Damage to Rupture of Quasi-Brittle Materials: Application to Seismic Faults*	coPI
2011	*National Science Foundation (instrumentation) , USA* *"Optical Interferometer Assembly for Time-Resolved Pointwise Stress & Velocity Measurements in Laboratory Earthquakes"*	coPI
2011	*Southern California Earthquake Center (SCEC) , USA* *"The Effect of Off-Fault Damage on the Propagation of an Earthquake Rupture"*	coPI
2010	*National Earthquake Hazards Reduction Program (USGS) , USA* *"Geometric Heterogeneities in Rupture Propagation for the Kunlun–Kunlun Pass–Xidatan Fault System"*	PI
2010	*National Science Foundation , USA* *"A Rate-Dependent Micromechanical Damage Mechanics for Simulating Dynamic Earthquake Ruptures"*	PI
2010	*National Nuclear Security Administration / U.S. Air Force , USA* *"Generation of High-Frequency P and S Energy by Rock Fracture During a Buried Explosion"*	coPI
2008	*National Science Foundation , USA* *Laboratory Earthquakes: Characterization of Ground Motion and Stress States in Complex Rupture Scenarios Using High-Resolution Optical Diagnostics*	coPI
2008	*National Science Foundation , USA* *"Micromechanics-Based Modeling of Dynamic Earthquake Rupture in a Structurally Complex Fault Zone"*	coPI

HONORS & AWARDS

2018 Grand Prix Michel Gouilloud Schlumberger — French Academy of Sciences
2018 CNRS Award for Doctoral Supervision and Research
2003 – 2006 Harvard University Certificate of Distinction in Teaching (2003, 2004, 2006)

STUDENTS

∗ → *current students*

Postdoctoral Associates

2026 – 2028	Mingqi Liu (China)∗	
2024 – 2026	Ankit Gupta (India)∗	
2025 – 2027	Suli Yao (China)∗	
2022 – 2025	Navid Kheirdast (Iran)	Post-Doctoral Fellow, ISTeP
2021 – 2023	Michelle Almakari (France)	
2021 – 2023	Carlos D. Villafuerte (Mexico)	Asst. Prof., UNAM
2019 – 2022	Ekeabino Momoh (Nigeria)	AXA Postdoc Fellow
2018 – 2021	Lucile Bruhat (France)	Nat. Catastrophe Risk Analyst, AXA

2019 – 2019	Lisa Gordeliy (Russia)	Senior Engineer, ResFrac
2014 – 2016	Marion Y. Thomas (France)	CNRS Scientist, Univ. de Rennes

PhD Students

2024 – 2027	Bharath Shanmugasundaram (India)*	
2024 – 2027	Yishuo Zhou (China)*	
2023 – 2026	Thomas Melkior (France)*	
2023 – 2027	Caiyuan Fan (China)*	
2021 – 2024	Jinhui Cheng (China)	Postdoc, Caltech
2020 – 2023	Augustin Thomas (France)	Research Scientist, BRGM
2020 – 2023	Joseph Flores Cuba (Peru)	Data Scientist, PowerBI
2018 – 2021	Claudia Hulbert (France)	CEO, Geolabe
2017 – 2020	Samson Marty (France)	Postdoc, Caltech
2014 – 2019	Marshall A. Martinez (USA)	Engineer, Joby Aviation
2015 – 2018	Kurama Okubo (Japan)	Research Scientist, NIED, Japan
2014 – 2017	Pierre Romanet (France)	Research Scientist, Cerema, France
2010 – 2015	Vahe Gabuchian (USA)	Research Scientist, Caltech
2011 – 2014	François X. Passelègue (France)	CNRS Scientist, GeoAzur, Nice
2008 – 2013	Jonathan Mihaly (USA)	Jet Propulsion Laboratory
2007 – 2012	Michael Mello (USA)	Teaching Professor, Caltech

Undergraduate & Masters Interns

2022	Valentin Marnat	2018	Nicolas Mercury	2013	Thibaut Perol
2021	Roxane Ferry	2018	Luc Illien	2012	Lucile Bruhat
2020	Jinhui Cheng	2017	Phillipe Danre	2004	Marion Olives
2019	Roxane Ferry	2015	Eleni Kolokytha	2003	Sonia Fliss
2019	Phillipe Danre	2015	Victor Barolle		
2019	Hugo Lestrelin	2014	Kurama Okubo		

TEACHING

Current — École Normale Supérieure

Earthquake Source Mechanics | Numerical Methods in Geosciences | Thermodynamics

Past — Harvard, Caltech, IPGP, ENS, École Polytechnique (with various colleagues)

Mécanique des Milieux Continus | Active Faults: Geometry | Seismic Ruptures and Scaling Laws | Introduction to Rock Physics | Mathematical Methods in the Sciences | Environmental Risks and Disasters | Ordinary and Partial Differential Equations | Complex and Fourier Analysis | Computational Solid and Structural Mechanics | Solid Mechanics | Introduction to the Mechanics of Solids | Mechanics of Fracture | Advanced Geomechanics | Mécanique de la Fracturation | Mécanique de la Rupture | Mécanique des Matériaux et des Structures

ORGANIZATION OF SCIENTIFIC MEETINGS

Jan 2026	Earthquake Source: Mechanics, Seismology and Geology Workshop — NISER, India
2023, 2024	Across the Time Scales, from Earthquakes to Earthquake Cycle — EGU
Jun 2019	Coupled Processes in Fracture Propagation in Geo-Materials — CISM Advanced School, Udine
Apr 2015	Multiscale Modeling of Fragmentation & Damage in Fault Zones — SSA

Dec 2014	Fault Zone Properties & Processes During Dynamic Ruptures — AGU

RESPONSIBILITIES & SERVICE

2018 – 2024	Team Leader, Faults & Earthquakes Group — ENS
2018 – 2019	Co-organizer, Internal Seminar — ENS
2025	Member, HCERES Committee for ISTerre — Univ. Grenoble
2024 – 2025	Editor, JTCAM

REVIEWING ACTIVITIES

American Geophysical Union | Seismological Society of America | Science | Nature | Nature Communications | Nature Geoscience | Science Advances | Proceedings of the National Academy of Sciences | Journal of the Mechanics and Physics of Solids | European Journal of Mechanics – A/Solids | International Journal of Fracture | Earth and Planetary Science Letters | Geophysical Research Letters | Journal of Structural Geology | Geology | Geophysical Journal International | Journal of Applied Mechanics | Geological Society of America | National Science Foundation | European Research Council

BOOKS

2017 Thomas, M. Y., T. M. Mitchell, and **H. S. Bhat**, eds. (2017). *Fault Zone Dynamic Processes : Evolution of Fault Properties During Seismic Rupture, Geophysical Monograph 227*. American Geophysical Union (AGU). DOI: 10.1002/9781119156895.

2012 Bizzarri, A. and **H. S. Bhat**, eds. (2012). *The mechanics of faulting: From laboratory to earthquakes*. Research Signpost.

BOOK CHAPTERS

2022 Thomas, M. Y. and **Bhat**, **H. S.** (2022a). “Loi de friction et modélisation numérique du cycle sismique”. in Le Cycle Sismique. Ed. by F. Rolandone. ISTE-Wiley. DOI: 10.51926/iste.9038.ch4.

2022 Thomas, M. Y. and **Bhat**, **H. S.** (2022b). “Friction Laws and Numerical Modeling of the Seismic Cycle”. in The Seismic Cycle: From Observation to Modeling. Ed. by F. Rolandone. ISTE-Wiley. DOI: 10.1002/9781394173709.ch4.

PUBLICATIONS

In Preparation

2026 Kheirdast, N., **Bhat**, **H. S.**, Almakari, M., Villafuerte, C., Thomas, M. Y., Gupta, A., and Dubernet, P. (2026). “Energy budget of spectrum of slip dynamics emerging from simplified model of fault and damage zone architecture”. **to be subm. J. Geophys. Res.**

2026 Garagash, D. I., Brantut, N., **Bhat**, **H. S.**, Schubnel, A., and Jolivet, R. (2026). “Low effective stress along faults caused by upwelling fluid flow in laboratory and nature”. **to be subm. J. Geophys. Res.**

2026 Gupta, A., Jara, J., Zhou, Y., Bhattacharya, P., Bollinger, L., and **Bhat**, **H. S.** (2026). “Effect of non-tectonic stress perturbations on rate-dependent frictional faults”. **to be subm. Geophys. Res. Lett.**

2026 Gupta, A., **Bhat**, **H. S.**, Faulkner, D. R., Bhattacharya, P., and Bollinger, L. (2026). “Response of stable intraplate fault zones to transient stress perturbations”. **to be subm. J. Geophys. Res.**

2026 Fan, C., Lin, G., Aubry, J., Deldicque, D., Giorgetti, C., **Bhat**, **H. S.**, and Schubnel, A. (2026). “Transition from thermal pressurization to dilatant strengthening during stick-slip ruptures in thermally cracked westerly granite”. **to be subm. J. Geophys. Res.**

2026 Momoh, E., Tait, S., and **Bhat**, **H. S.** (2026a). “3-D numerical modelling of the feedback between deformation and thermal structure during subduction initiation for the French Lesser Antilles”. **to be subm. Tectonics**.

2026 Momoh, E., Tait, S., and **Bhat**, **H. S.** (2026b). "3-D thermomechanical geodynamic modelling of the Massif Central (France) and Eifel Volcanic Region (Germany)". **to be subm. Tectonophysics**.

2026 Bhagat, R., Sreejith, K., Satriano, C., Gahalaut, V., **Bhat**, **H. S.**, and Bhattacharya, P. (2026). "Aseismic slip and low-frequency earthquakes within an intraplate seismic swarm: the role of fault network geometry". **to be subm.**

2026 Bhagat, R., Bhattacharya, P., Sreejith, K. M., **Bhat**, **H. S.**, and Gahalaut, V. K. (2026). "Coupled Transient Processes Govern Intraplate Earthquake Swarm Evolution: Insights from the 2019–2020 Palghar Sequence". **to be subm. AGU Advances**.

2026 Yao, S., Shanmugasundaram, B., Brantut, N., Jara, J., Aochi, H., Yang, H., and **Bhat**, **H. S.** (2026). "Inferring fault scale friction from a near fault station". **to be subm.**

Under Review

2026 Zhou, Y., Aochi, H., Schubnel, A., Ide, S., and **Bhat**, **H. S.** (2026). "Tidal sensitivity of tremors in a mixed fast and slow earthquake system in northeastern Japan". **subm. J. Geophys. Res.** DOI: `10.48550/arXiv.2606.24362`.

2026 Yao, S., Yang, H., **Bhat**, **H. S.**, and Aochi, H. (2026). "Supershear Rupture Indicator in Near-fault Particle Motion". **submitted**. DOI: `10.48550/arXiv.2606.10843`.

2026 Thomas, M. Y. and **Bhat**, **H. S.** (2026). "Une Histoire de la compréhension des Tremblements de Terre". in Histoire et philosophie des sciences. Ed. by Y. Vaills. ISTE.

2026 Cheng, J., **Bhat**, **H. S.**, Almakari, M., Lecampion, B., and Dubernet, P. (2026). "Quantifying the Role of 3D Fault Geometry Complexities on Slow and Fast Earthquakes". **to appear in Geophys. Res. Lett.** DOI: `10.48550/arxiv.2602.16403`.

2026 Zhou, Y., Gupta, A., Aochi, H., Schubnel, A., Ide, S., and **Bhat**, **H. S.** (2026). "Theoretical Constraints on Tidal Triggering of Slow Earthquakes". **minor revisions J. Geophys. Res.** DOI: `10.48550/arxiv.2602.06703`.

2026 Wang, C., Wang, P., Wu, B., **Bhat**, **H. S.**, Bhattacharya, P., Xie, Y., Xia, K., and Schubnel, A. (2026). "Complex fault slip behavior modulated by fluid injection rate". **submitted**.

Published

2026 Almakari, M., Kheirdast, N., Villafuerte, C. D., Thomas, M. Y., Dubernet, P., Cheng, J., Gupta, A., Romanet, P., Chaillat, S., and **Bhat**, **H. S.** (2026). "Fault volume digital twin to reproduce the full slip spectrum, scaling and statistical laws". **J. Geophys. Res.** 131, e2025JB032915. DOI: `10.1029/2025JB032915`.

2026 Melkior, T., **Bhat**, **H. S.**, and Amlani, F. (2026). "Tsunami modeling with dynamic seafloors: a high-order solver validated with shallow water benchmarks". **J. Comput. Phys.** 562.114990. DOI: `10.1016/j.jcp.2026.114990`.

2026 Michel, S., Scotti, O., Hok, S., **Bhat**, **H. S.**, Kheirdast, N., Romanet, P., Almakari, M., and Cheng, J. (2026). "A rate-and-state friction based criterion for the probability of earthquake fault jumps". **J. Geophys. Res.** 131, e2025JB031449. DOI: `10.1029/2025JB031449`.

2025 Latour, S., Lebihain, M., **Bhat**, **H. S.**, Twardzik, C., Bletery, Q., Hudnut, K. W., and Passelègue, F. (2025). "Direct Estimation of Earthquake Source Properties from a Single CCTV Camera". **Science** 390, pp. 463–467. DOI: `10.1126/science.adz1705`.

2025 Cheng, J., Almakari, M., Peruzzo, C., Lecampion, B., and **Bhat**, **H. S.** (2025). "FASTDASH, a Quasi-dynamic 3D Seismic Cycle Model by Using Boundary Element Method with H-matrices". **Geophys. J. Int.** DOI: `10.1093/gji/ggaf230`.

2025 Momoh, E., **Bhat**, **H. S.**, Tait, S., and Gerbault, M. (2025). "Volumetric (dilatant) plasticity in geodynamic models and implications on thermal dissipation and strain localization". **Geophys. J. Int.** 240.3, pp. 1551–1578. DOI: `10.1093/gji/ggae463`.

2025 Ferry, R., Thomas, M. Y., **Bhat**, **H. S.**, and Dubernet, P. (2025). "Depth Dependence of Coseismic Off-Fault Damage and its Effects on Rupture Dynamics". **J. Geophys. Res.** e2024JB029787. DOI: `10.1029/2024jb029787`.

2024 Petit, L., Olive, J.-A., Schubnel, A., Le Pourhiet, L., and **Bhat**, **H. S.** (2024). "A brittle constitutive law for long-term tectonic modeling based on sub-critical crack growth". **to appear in Geochem. Geophys. Geosyst.** 25, e2023GC011229. DOI: `10.1029/2023gc011229`.

2023 Jeandet-Ribes, L., Thomas, M. Y., and **Bhat**, **H. S.** (2023). “On the importance of setting 3-D stress field in simulations of on- and off-fault deformation”. **Geophys. J. Int.** 235.3, pp. 2962–2978. DOI: 10.1093/gji/ggad401.

2023 Marty, S., Schubnel, A., **Bhat**, **H. S.**, Aubry, J., Fukuyama, E., Latour, S., Nielsen, S., and Madariaga, R. (2023). “Nucleation of laboratory earthquakes: quantitative analysis and scalings”. **J. Geophys. Res.** 128.e2022JB026294. DOI: 10.1029/2022jb026294.

2022 Amlani, F., **Bhat**, **H. S.**, Simons, W. J. F., Schubnel, A., Vigny, C., Rosakis, A. J., Efendi, J., Elbanna, A., Dubernet, P., and Abidin, H. Z. (2022). “Supershear shock front contribution to the tsunami from the 2018 Mw 7.5 Palu, Indonesia earthquake”. **Geophys. J. Int.** 230, pp. 2089–2097. DOI: 10.1093/gji/ggac162.

2021 Jara, J., Bruhat, L., Thomas, M. Y., Antoine, S., Okubo, K., Klinger, Y., Jolivet, R., and **Bhat**, **H. S.** (2021). “Signature of transition to supershear rupture speed in coseismic off-fault damage zone”. **Proc. R. Soc. A.** 477, p. 20210364. DOI: 10.1098/rspa.2021.0364.

2021 Elbanna, A., Abdelmeguid, M., Ma, X., Amlani, F., **Bhat**, **H. S.**, Synolakis, C., and Rosakis, A. J. (2021). “Anatomy of Strike Slip Fault Tsunami Genesis”. **Proc. Natl. Acad. Sci. USA**. DOI: 10.1073/pnas.2025632118.

2021 **Bhat**, **H. S.** (2021). “Supershear Earthquakes”. *Habilitation à Diriger des Recherches, Ecole Normale Supérieure*.

2020 Jeandet-Ribes, L., Cubas, N., **Bhat**, **H. S.**, and Steer, P. (2020). “Response of a single fault to transient normal stress change, and implications of large erosional events on the seismic cycle”. **Geophys. Res. Lett.** 47.e2020GL087631. DOI: 10.1029/2020gl087631.

2020 Jolivet, R., Simons, M., Duputel, Z., Olive, J.-A., **Bhat**, **H. S.**, and Bletery, Q. (2020). “Interseismic Loading of Subduction Megathrust Drives Long-Term Uplift in Northern Chile”. **Geophys. Res. Lett.** 47.8, e2019GL085377. DOI: 10.1029/2019gl085377.

2020 Okubo, K., Rougier, E., Lei, Z., and **Bhat**, **H. S.** (2020). “Modeling earthquakes with off-fault damage using the combined finite discrete element method”. **J. Comp. Part. Mech.** DOI: 10.1007/s40571-020-00335-4.

2019 Okubo, K., **Bhat**, **H. S.**, Rougier, E., Marty, S., Schubnel, A., Lei, Z., Knight, E. E., and Klinger, Y. (2019). “Dynamics, radiation and overall energy budget of earthquake rupture with coseismic off-fault damage”. **J. Geophys. Res.** 124. DOI: 10.1029/2019jb017304.

2019 Marty, S., Passelègue, F. X., Aubry, J., Schubnel, A., **Bhat**, **H. S.**, and Madariaga, R. (2019). “Origin of high-frequency radiation during laboratory earthquakes”. **Geophys. Res. Lett.** 46. DOI: 10.1029/2018gl080519.

2018 Aubry, J., Passelègue, F. X., Deldicque, D., Girault, F., Marty, S., Lahfid, A., **Bhat**, **H. S.**, Escartin, J., and Schubnel, A. (2018). “Frictional heating processes and energy budget during laboratory earthquakes”. **Geophys. Res. Lett.** 45. DOI: 10.1029/2018gl079263.

2018 Klinger, Y., Okubo, K., Vallage, A., Champenois, J., Delorme, A., Rougier, E., Lei, Z., Knight, E. E., Munjiza, A., Satriano, C., Baize, S., Langridge, R., and **Bhat**, **H. S.** (2018). “Earthquake damage patterns resolve complex rupture processes”. **Geophys. Res. Lett.** DOI: 10.1029/2018gl078842.

2018 Cruz-Atienza, V. M., Villafuerte, C. D., and **Bhat**, **H. S.** (2018). “Rapid tremor migration and pore-pressure waves in subduction zones”. **Nat. Commun.** 9.1, p. 2900. DOI: 10.1038/s41467-018-05150-3.

2018 Thomas, M. Y. and **Bhat**, **H. S.** (2018). “Dynamic evolution of off-fault medium during an earthquake: a micromechanics based model”. **Geophys. J. Int.** 214.2, pp. 1267–1280. DOI: 10.1093/gji/ggy129.

2018 Romanet, P., **Bhat**, **H. S.**, Jolivet, R., and Madariaga, R. (2018). “Fast and slow earthquakes emerge due to fault geometrical complexity”. **Geophys. Res. Lett.** DOI: 10.1029/2018gl077579.

2017 Gabuchian, V., Rosakis, A. J., **Bhat**, **H. S.**, Madariaga, R., and Kanamori, H. (2017). “Experimental evidence that thrust earthquake ruptures might open faults”. **Nature** 545.336–339. DOI: 10.1038/nature22045.

2017 Thomas, M. Y., **Bhat**, **H. S.**, and Klinger, Y. (2017). “Effect of Brittle off-fault Damage on Earthquake Rupture Dynamics”. in Fault Zone Dynamic Processes : Evolution of Fault Properties During Seismic Rupture, Geophysical Monograph 227. Ed. by M. Y. Thomas, **H. S. Bhat**, and T. M. Mitchell. American Geophysical Union (AGU), pp. 255–280. DOI: 10.1002/9781119156895.ch14.

2017 Passelègue, F. X., Latour, S., Schubnel, A., Nielsen, S., **Bhat**, **H. S.**, and Madariaga, R. (2017). “Precursory Processes during Laboratory Earthquakes”. in Fault Zone Dynamic Processes : Evolution of Fault Properties During Seismic Rupture, Geophysical Monograph 227. Ed. by M. Y. Thomas, **H. S. Bhat**, and

T. M. Mitchell. American Geophysical Union (AGU). Chap. 12, pp. 229–242. DOI: 10.1002/9781119156895.ch12.

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